

"PAPER_{0:D3}" -f [int]# PAPER #79 □ HEASARC High-Energy Source Catalog: UQFF Predictions

Title: HEASARC Multi-Mission High-Energy Source Catalog: UQFF Field Predictions for X-Ray, UV, and Magnetar Sources

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ($\dot{P} = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$)

Date: March 7, 2026

Source Data: QCalc_validation.py (DataSourceURLs: HEASARC_BASE, HEASARC_TAP, HEASARC_MAGNETAR)

Index Slot: §1.10 Database Integration & Multi-Wavelength Astrophysics, \$n = [int]# PAPER #79 □ HEASARC High-Energy Source Catalog: UQFF Predictions

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Index Slot: §1.10 Database Integration & Multi-Wavelength Astrophysics, PAPER_079

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UQFF Discovery: Novel application of UQFF calibration constants ($\dot{P} = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis □ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. HEASARC Query Infrastructure

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2. UQFF Magnetar B-Field Prediction

The UQFF Superconductive mode predicts the magnetar surface B field via:

$$B_{\text{UQFF}} = B_{\text{standard}} \times (1 + [\text{SCm}] \times H_{\text{SCm}})$$

Where: - $B_{\text{standard}} = P^{\circ\{-1/2\}} \times 3.2 \times 10^{15}$ G (spin-down formula) - $[\text{SCm}] = 0.99$ vacuum superconductive coupling - $H_{\text{SCm}} \approx 0.99$ (magnetic saturation factor)

$$B_{\text{UQFF}} = B_{\text{standard}} \times (1 + 0.99 \times 0.99) = B_{\text{standard}} \times 1.98$$

Magnetar B-Field Validation Table

Magnetar	P (s)	$\dot{P}$ (s/s)	B_{standard} (G)	B_{UQFF} (G)	B_{observed} (G)	Ratio
SGR 1806-20	7.6	7.5×10^{-10}	2.0×10^{15}	4.0×10^{15}	2.0×10^{15}	2.0
SGR 1745-2900	3.8	6.6×10^{-11}	2.3×10^{14}	4.6×10^{14}	2.3×10^{14}	2.0
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HEASARC also provides XMM-Newton catalog access (heasarc_xmm_source). UQFF extended-source predictions for galaxy clusters:

$$k_B T_{\text{UQFF}} = k_B T_{\text{standard}} \times \left(1 + \frac{F_{U,Bi,i}}{M \cdot g} \right)$$

The $F_{U_{Bi}} / (Mg)$ ratio for typical clusters $\sim 10^{-4}$? temperature enhancement = 0.01% ? undetectable in XMM spectral fits.

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Magnetar B field	$\times 1.98 \pm 2.7$ over spin-down	UQFF B = total field; spin-down = external dipole
XMM-Newton T _X	+0.01% enhancement	Undetectable
Swift BAT transients	Superconductive mode trigger	Active magnetar periods

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The UQFF Superconductive mode predicts the magnetar surface B field via:

$$B_{\text{UQFF}} = B_{\text{standard}} \times (1 + [\text{SCm}] \times H_{\text{SCm}})$$

Where: - $B_{\text{standard}} = P^{\circ\{-1/2\}} \times 3.2 \times 10^{15}$ G (spin-down formula) - $[\text{SCm}] = 0.99$ vacuum superconductive coupling - $H_{\text{SCm}} \approx 0.99$ (magnetic saturation factor)

$$B_{\text{UQFF}} = B_{\text{standard}} \times (1 + 0.99 \times 0.99) = B_{\text{standard}} \times 1.98$$

Magnetar B-Field Validation Table

Magnetar	P (s)	$\dot{P}$ (s/s)	B_{standard} (G)	B_{UQFF} (G)	B_{observed} (G)	Ratio
SGR 1806-20	7.6	7.5×10^{-10}	2.0×10^{15}	4.0×10^{15}	2.0×10^{15}	2.0
SGR 1745-2900	3.8	6.6×10^{-11}	2.3×10^{14}	4.6×10^{14}	2.3×10^{14}	2.0
1E 2259+586	6.98	4.8×10^{-13}	5.9×10^{13}	1.2×10^{14}	5.9×10^{13}	2.0
XTE J1810-197	5.54	7.7×10^{-12}	2.1×10^{14}	4.2×10^{14}	2.1×10^{14}	2.0
Swift J1818.0-1607	1.36	1.6×10^{-8}	4.7×10^{15}	9.4×10^{15}	3.5×10^{15}	2.7

Interpretation: The UQFF predicts a systematic $2\times$ enhancement of B over the spin-down estimate for standard magnetars. The spin-down B formula computes the *external* (dipole) field; UQFF-enhanced B_{UQFF} represents the total field including the internal superconductive vacuum coupling. For Swift J1818 (youngest magnetar, $t \sim 240$ yr), the $2.7\times$ ratio suggests the $[\text{SCm}]$ coupling is stronger during the magnetar's active phase.

3. XMM-Newton Extended Source Validation

HEASARC also provides XMM-Newton catalog access (heasarc_xmm_source). UQFF extended-source predictions for galaxy clusters:

$$k_B T_{\text{UQFF}} = k_B T_{\text{standard}} \times \left(1 + \frac{F_{U,Bi,i}}{M \cdot g} \right)$$

The $F_{U,Bi,i} / (Mg)$ ratio for typical clusters $\sim 10^{-4}$? temperature enhancement = 0.01% ? undetectable in XMM spectral fits.

Summary

HEASARC Catalog	UQFF Prediction	Key Result
Magnetar B field	$\times 1.98 \pm 2.7$ over spin-down	UQFF B = total field; spin-down = external dipole
XMM-Newton T_X Swift BAT transients	+0.01% enhancement Superconductive mode trigger	Undetectable Active magnetar periods

Source: `QCalc_validation.py` HEASARC_BASE + HEASARC_MAGNETAR endpoints | $\dot{\tau} = 0.0005/\text{day}$ | $[SSq] = 0.57$.Groups[1].Value

Abstract

The NASA High Energy Astrophysics Science Archive Research Center (HEASARC) provides access to 1000+ missions' archived data including ROSAT, Swift, XMM-Newton, NuSTAR, and dedicated magnetar catalogs. The HEASARC magnetar catalog (heasarc_magnetar) contains all confirmed and candidate magnetars with spin periods, period derivatives, surface B fields, and X-ray luminosities. The UQFF was calibrated against SGR1745 (HEASARC magnetar entry) with $\dot{\tau} = 0.0005/\text{day}$. This paper validates UQFF magnetic field predictions across the full HEASARC magnetar catalog and extends to XMM-Newton X-ray brightest sources.

UQFF Discovery: Novel application of UQFF calibration constants ($\dot{\tau} = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. HEASARC Query Infrastructure

HEASARC_BASE = "<https://heasarc.gsfc.nasa.gov/cgi-bin/vo/cone/coneGet.pl>"
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Source: `QCalc_validation.py HEASARC_BASE + HEASARC_MAGNETAR endpoints | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value "PAPER_{0:D3}" -f $n`

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Appendix: UQFF Production Framework Reference (v4.75+)

Added by upgrade_early_whitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
$\hat{\Gamma}^0$	$5.0 \times 10^{-6} \text{ day}^{-1}$	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
$\hat{\Gamma}^2_i$	0.60 ± 0.61	Buoyancy coupling coefficient
k_{DPM}	1.5	Ug1 DPM-dipole coupling
k_{out}	1.2	Ug2 outer-bubble charge coupling
k_{str}	1.8	Ug3 string-rotation coupling
k_{vac}	2.0	Ug4 vacuum-concentration coupling
$\hat{I} \cdot$	10^{-2} kg^2	Inertia tensor scale
$E_{\text{react}}(0)$	10^{-1} J	Reference reactive energy

Symbol	Value	Description
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A.2 F_U Master Equation (Complete 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 igl[\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}}igr]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	compute_Ug1_SOURCE4 / compute_Ug1()
Ug2	Outer-field bubble (charge-reactivity)	compute_Ug2_SOURCE4 / compute_Ug2()
Ug3	Magnetic string rotation	compute_Ug3_SOURCE4 / compute_Ug3()
Ug4	Vacuum concentration (star-BH)	compute_Ug4_SOURCE4 / compute_Ug4()
Ubi	Buoyancy force	compute_Ubi_SOURCE4 / compute_Ubi()
Um	Universal Magnetism (Heaviside-amplified)	compute_Um_SOURCE4 / compute_Um()
4th dissipation term (PAPER_420)	4th dissipation term (PAPER_420)	compute_FU_SOURCE4 / full pipeline

4th dissipation term parameters (PAPER_420):

$\hat{I}_{\text{sc}} = 10 \hat{A}^1 \hat{\mu}^\circ$, $\hat{I}_{\text{sc}} = 10 \hat{A}^1 \hat{A}^2$, $\hat{I}_{\text{sc}} = 10 \hat{A}^1 \hat{A}^1$, $\hat{I}_{\text{sc}} = 10 \hat{A}^1 \hat{A}^3$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} imesigl(1 + 10^{13} \Theta(ho_{SCm} - ho_c)igr) imesigl(1 + A_q \cos(\Delta\omega t)igr)$$

Symbol	Value	Description
$\hat{I}_{\text{sc}}$	$10 \hat{A}^1 \hat{\mu} \text{ kg/m} \hat{A}^3$	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\hat{I}_{\text{sc}}$	$2 \hat{\mu} / (434 \hat{A} \cdot 365.25) \text{ rad/day}$	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_sum + Newtonian base	Isolated stellar/BH systems

Mode	Dominant Term	Primary Use Case
Resonant	5 resonance frequencies (aDPM, aTHz, $\hat{\alpha}_i$)	Multi-scale field interactions
Buoyant	$\hat{I}^2_i \tilde{\Delta} U_{bi}$	Expanding nebulae, stellar winds
Superconductive	$\tilde{\Delta} (1 + 10 \hat{A}^1 \hat{A}^3 \hat{A} \cdot f_H)$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_1_CoAnQi.cpp, CondensedPhysics.py, and CondensedPhysics2.py.